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Below is a **lecture scribe written strictly and exclusively from the provided lecture slides** for **CSE400 – Lecture 8**.
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The scribe is intended to function as a **standalone, exam-oriented revision document** for a closed-notes or reading-based exam.

CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables

Instructor: Dhaval Patel, PhD
Date: January 29, 2026

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1. Lecture Outline (As Presented)

The lecture covers the following topics in order:

1. Gaussian Random Variable
    - Definition and Properties

    - Standard Forms: Q-function and Error Function
    - Relation between Φ-function and Q-function
    - Example
2. Types of Continuous Random Variable
    - Uniform Random Variable: Example
    - Exponential Random Variable: Example
    - Laplace Random Variable: Example
    - Gamma Random Variable
        - Graph and Special Cases
        - Example
3. Homework Problem
4. Problem Solving
5. In-Class Activity: Gaussian Density Estimation

---

2. Gaussian Random Variable

2.1 Definition

A **Gaussian random variable** is defined as a random variable whose **probability density function (PDF)** can be written in the following general form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

where:

- m is the mean

- σ^2 is the variance

This definition is explicitly given on the slide titled **"Gaussian Random Variable – Definition and Properties"**

2.2 Interpretation Using PDF and CDF (As Shown)

The lecture slide visually presents:

- The **PDF** of a Gaussian random variable
- The **CDF**, which is the integral of the PDF

A figure illustrates:

- The bell-shaped PDF
- The corresponding CDF curve
- An example Gaussian with $m = 3$ and $\sigma = 2$

No additional interpretation beyond the graphical illustration is provided

---

3. Standard Forms Associated with Gaussian RV

3.1 Error Function

The **error function** is defined as:

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x \exp(-t^2)\, dt$$

This definition is given explicitly on the slide

3.2 Complementary Error Function

The **complementary error function** is defined as:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty \exp(-t^2)\, dt$$

This relationship is stated exactly as shown, with no additional derivation

---

3.3 Φ-function (CDF of Standard Normal)

The **Φ-function** is defined as the **CDF of a standard normal random variable**:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(-\frac{t^2}{2}\right)\, dt$$

This definition is presented under “Standard Forms”

3.4 Q-function (Gaussian Tail Function)

The **Q-function** is defined as:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right)\, dt$$

This is labeled explicitly as the **Gaussian Tail Function**

3.5 Relation Between Φ-function and Q-function

A note is given on the slide:

$$Q(x) = 1 - \Phi(x)$$

No further explanation is provided beyond this identity

---

4. Relation Between Φ-function and Q-function for a Gaussian RV

4.1 Evaluating the CDF

For a Gaussian random variable $X \sim \mathcal{N}(m, \sigma^2)$, the CDF is evaluated as:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

By change of variables shown on the slide:

$$= \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

Hence:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

This derivation is shown step by step on the slide without additional justification

---

4.2 Evaluating Tail Probabilities

The tail probability is evaluated as:

$$\Pr(X > x) = \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

Which gives:

$$\Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

This is explicitly written on the slide

4.3 Key Identities

The lecture lists the following identities:

1.

Q(x) = 1 - \Phi(x)

]

2.

F_X(x) = 1 - Q\left(\frac{x - m}{\sigma} \right)

]

These identities are presented as bullet points

---

5. Gaussian Random Variable – Worked Example

5.1 Given PDF

A random variable X has the PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

5.2 Identifying Parameters

Comparing with the Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

We identify:

- $m = -3$
- $\sigma^2 = 4$
- $\sigma = 2$

These values are explicitly stated on the solution slide

5.3 Gaussian Relations Used

The following relations are stated:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$\Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

5.4 Probability Calculations

(1) $\Pr(X \leq 0)$

$$\Pr(X \leq 0) = F_X(0)$$

$$= \Phi\left(\frac{0-(-3)}{2}\right)$$

$$= \Phi\left(\frac{3}{2}\right)$$

$$= 1 - Q\left(\frac{3}{2}\right)$$

---

(2) $\Pr(X > 4)$

$$\Pr(X > 4) = Q\left(\frac{4-(-3)}{2}\right)$$

$$= Q\left(\frac{7}{2}\right)$$

These results are presented exactly as shown on the slides, without numerical approximation

---

6. Notes Provided in the Lecture

- To evaluate the **CDF of a Gaussian random variable**, evaluate the **Φ-function** at the required points.
- The **Q-function** is more natural for evaluating probabilities of the form $\Pr(X > x)$.

These notes are included verbatim in highlighted boxes on the slides

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End of Lecture Scribe

This document strictly mirrors the lecture slides in:

- Content
- Order
- Level of detail

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CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables

Instructor: Dhaval Patel, PhD
Date: January 29, 2026

---

1. Lecture Outline (As Given in Slides)

The lecture is organized as follows:

1. Gaussian Random Variable
    - Definition and Properties
    - Standard Forms: Q-function and Error Function
    - Relation between Φ-function and Q-function

    - Example
2. Types of Continuous Random Variable
    - Uniform Random Variable: Example
    - Exponential Random Variable: Example
    - Laplace Random Variable: Example
    - Gamma Random Variable
        - Graph and Special Cases
        - Example
3. Homework Problem
4. Problem Solving
5. In-Class Activity: Gaussian Density Estimation

> **Note:** In the provided slides, **only the Gaussian-related material and general expectation/moment material are explicitly shown in detail**. Other topics are listed in the outline but **do not appear with definitions or examples in the provided slides**, and therefore are **not expanded further** in this scribe.

---

2. Moments and Expectations (Preliminary Material)

2.1 Expected Value of a Function of a Random Variable

Continuous Random Variable (CRV)
The expected value of a function $g(X)$ is defined as:

$$\mathbb{E}[g(X)] = \int g(x)\, f_X(x)\, dx$$

Discrete Random Variable (DRV)

The expected value of a function $g(X)$ is defined as:

$$\mathbb{E}[g(X)] = \sum_k g(x_k)\, P_X(x_k)$$

These formulas are written explicitly on the slide titled **"Expected value of g of RV: X"**

2.2 Linearity Properties of Expectation

For any constants a and b:

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$$

For a sum of functions:

$$\mathbb{E}\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} \mathbb{E}[g_k(X)]$$

These properties are stated directly on the slide without proof

---

3. Higher-Order Moments

3.1 Third Central Moment – Skewness

The third central moment is defined as:

$$m_3 = C_s = \frac{\mathbb{E}[(X - \mu_X)^3]}{\sigma_X^3}$$

Interpretation as written on the slide:

- It is a measure of **symmetry of the PDF about the mean** μ_X
- Positive value → right side
- Negative value → left side

No further explanation or derivation is provided

3.2 Fourth Central Moment – Kurtosis

The fourth central moment is defined as:

$$m_4 = C_k = \frac{\mathbb{E}[(X - \mu_X)^4]}{\sigma_X^4}$$

Interpretation as written on the slide:

- Large value → the random variable has a **large peak near the mean**

A sketch illustrating a high peak is shown, with no additional mathematical explanation

---

4. Gaussian Random Variable

4.1 Definition

A **Gaussian random variable** is defined as a random variable whose **PDF** can be written in the general form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

This definition appears under "**Gaussian Random Variable – Definition and Properties**"

4.2 PDF and CDF Illustration

The slides show:

- A plot of the **PDF**
- A plot of the **CDF**
- A labeled example with $m = 3$ and $\sigma = 2$

The relationship shown graphically:

- PDF integrates to CDF
- $F_X(x) + Q(x) = 1$

No algebraic explanation beyond what is drawn is included

---

5. Standard Forms for Gaussian Random Variables

5.1 Error Function

The **error function** is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

5.2 Complementary Error Function

The **complementary error function** is defined as:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt$$

These definitions are shown exactly as written on the slides

5.3 Φ-function (CDF of Standard Normal)

The **Φ-function** is defined as:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(-\frac{t^2}{2}\right) dt$$

It is explicitly labeled as the **CDF of the standard normal**

5.4 Q-function (Gaussian Tail Function)

The **Q-function** is defined as:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt$$

It is explicitly labeled as the **Gaussian Tail Function**

5.5 Relationship Between Φ and Q

A note on the slide states:

$$Q(x) = 1 - \Phi(x)$$

No derivation is provided

---

6. Relation Between Φ-function and Q-function for a Gaussian RV

6.1 Evaluating the CDF

For $X \sim \mathcal{N}(m, \sigma^2)$:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Change of variables shown on the slide:

$$= \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

Hence:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

All steps appear exactly as written

6.2 Evaluating Tail Probabilities

$$\Pr(X > x) = \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$
$$= Q\left(\frac{x-m}{\sigma}\right)$$

6.3 Key Identities

The following identities are listed:

$$Q(x) = 1 - \Phi(x)$$
$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

6.4 Notes from Slides

- To evaluate the **CDF** of a Gaussian RV, evaluate the **Φ-function** at the points.
- The **Q-function** is more natural for evaluating probabilities of the form $\Pr(X > x)$.

---

7. Gaussian Random Variable – Worked Example

7.1 Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

7.2 Identification of Parameters

Comparing with:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

We identify:

$$m = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

7.3 Gaussian Relations Used

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$\Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

7.4 Probability Calculations

(1) $\Pr(X \leq 0)$

$$\Pr(X \leq 0) = F_X(0)$$
$$= \Phi\left(\frac{0-(-3)}{2}\right) = \Phi\left(\frac{3}{2}\right)$$
$$= 1 - Q\left(\frac{3}{2}\right)$$

---

(2) $\Pr(X > 4)$

$$\Pr(X > 4) = Q\left(\frac{4-(-3)}{2}\right) = Q\left(\frac{7}{2}\right)$$

All steps and results are exactly as shown on the slides

End of Lecture Scribe

This scribe:

- Uses **only** the provided slides
- Preserves **exact order, notation, and emphasis**
- Includes **no additional theory or assumptions**

It is suitable for **direct exam revision** without consulting any other source.

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CSE400 – Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables

Instructor: Dhaval Patel, PhD

Date: January 29, 2026

---

1. Lecture Outline (As Shown on Slides)

The lecture is organized into the following sections:

1. **Gaussian Random Variable**
    - Definition and Properties
    - Standard Forms: Q-function and Error Function
    - Relation between Φ-function and Q-function
    - Example
2. **Types of Continuous Random Variable**

- Uniform Random Variable: Example
- Exponential Random Variable: Example
- Laplace Random Variable: Example
- Gamma Random Variable
  - Graph and Special Cases
  - Example

3. Homework Problem
4. Problem Solving
5. In-Class Activity: Gaussian Density Estimation

> Only the topics that are **explicitly developed on the slides** are expanded below.
> Topics listed in the outline but **not developed further on the slides** are not expanded.

---

2. Expected Value and Moments (Material Appearing Before Gaussian RV)

2.1 Expected Value of a Function of a Random Variable

Continuous Random Variable (CRV)
The expected value of a function $g(X)$ is given by:

$$\mathbb{E}[g(X)] = \int g(x)\, f_X(x)\, dx$$

Discrete Random Variable (DRV)

The expected value of a function $g(X)$ is given by:

$$\mathbb{E}[g(X)] = \sum_k g(x_k)\, P_X(x_k)$$

These formulas are written explicitly on the slide titled
"Expected value of g of RV: X"

2.2 Linearity Properties of Expectation

For constants a and b:

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$$

For a finite sum of functions:

$$\mathbb{E}\left[\sum_{k=1}^{N} g_k(X)\right] = \sum_{k=1}^{N} \mathbb{E}[g_k(X)]$$

These properties are stated directly on the slide, with no proof or additional explanation

---

3. Higher-Order Central Moments

3.1 Third Central Moment – Skewness

The third central moment is defined as:

$$m_3 = C_s = \frac{\mathbb{E}[(X - \mu_X)^3]}{\sigma_X^3}$$

Explanation exactly as written on the slide:

- It is a **measure of symmetry of the PDF about the mean** μ_X
- Positive value → **Right**
- Negative value → **Left**

No additional explanation is given

3.2 Fourth Central Moment – Kurtosis

The fourth central moment is defined as:

$$m_4 = C_k = \frac{\mathbb{E}[(X - \mu_X)^4]}{\sigma_X^4}$$

Explanation exactly as written on the slide:

- Large value → the random variable will have a **large peak near the mean**

A sketch of a sharp peak is shown. No algebraic derivation is provided

---

4. Gaussian Random Variable

4.1 Definition

A **Gaussian random variable** is defined as a random variable whose **PDF** can be written in the general form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

This definition appears under
"Gaussian Random Variable – Definition and Properties"

4.2 PDF and CDF Representation

The slide shows:

- The **PDF** of a Gaussian random variable
- The corresponding **CDF**
- A labeled example with $m = 3$ and $\sigma = 2$

The visual relationship shown:

- Area under the PDF corresponds to the CDF
- Left-tail area corresponds to $\Phi(x)$
- Right-tail area corresponds to $Q(x)$

No algebraic explanation beyond the figure is included

---

5. Standard Forms for Gaussian Random Variables

5.1 Error Function

The **error function** is defined as:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

5.2 Complementary Error Function

The **complementary error function** is defined as:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2}\, dt$$

Both definitions are written explicitly on the slides

5.3 Φ-Function (CDF of Standard Normal)

The **Φ-function** is defined as:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x \exp\left(-\frac{t^2}{2}\right) dt$$

It is explicitly labeled as the **CDF of the Standard Normal**

5.4 Q-Function (Gaussian Tail Function)

The **Q-function** is defined as:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt$$

It is explicitly labeled as the **Gaussian Tail Function**

5.5 Relationship Between Φ and Q

A note on the slide states:

$$Q(x) = 1 - \Phi(x)$$

No derivation is provided

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6. Relation Between Φ-Function and Q-Function for a Gaussian RV

6.1 Evaluating the CDF

For a Gaussian random variable $X \sim \mathcal{N}(m, \sigma^2)$:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Change of variable shown on the slide:

$$= \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

Therefore:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

All steps appear explicitly on the slide

6.2 Evaluating Tail Probabilities

$$\begin{aligned}\Pr(X > x) &= \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt \\ &= Q\left(\frac{x-m}{\sigma}\right)\end{aligned}$$

6.3 Key Identities

The following identities are listed:

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

6.4 Notes Provided on Slides

- To evaluate the **CDF** of a Gaussian random variable, evaluate the **Φ-function** at the points.

- The **Q-function** is more natural for evaluating probabilities of the form $\Pr(X > x)$.

---

7. Gaussian Random Variable – Worked Example

7.1 Given PDF

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

7.2 Identification of Parameters

Comparing with the Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

We identify:

$$m = -3, \quad \sigma^2 = 4, \quad \sigma = 2$$

7.3 Gaussian Relations Used

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$\Pr(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

7.4 Probability Calculations

(1) $\Pr(X \leq 0)$

$$\Pr(X \leq 0) = F_X(0)$$

$$= \Phi\left(\frac{0 - (-3)}{2}\right) = \Phi\left(\frac{3}{2}\right)$$

$$= 1 - Q\left(\frac{3}{2}\right)$$

---

(2) $\Pr(X > 4)$

$$\Pr(X > 4) = Q\left(\frac{4 - (-3)}{2}\right) = Q\left(\frac{7}{2}\right)$$

All steps are reproduced exactly as shown on the slides

End of Lecture Scribe

This document:

- Uses **only** the lecture slides as context

- Preserves **exact order, notation, and wording**
- Adds **no external explanations or assumptions**

It is suitable as a **primary exam-revision reference** for this lecture.